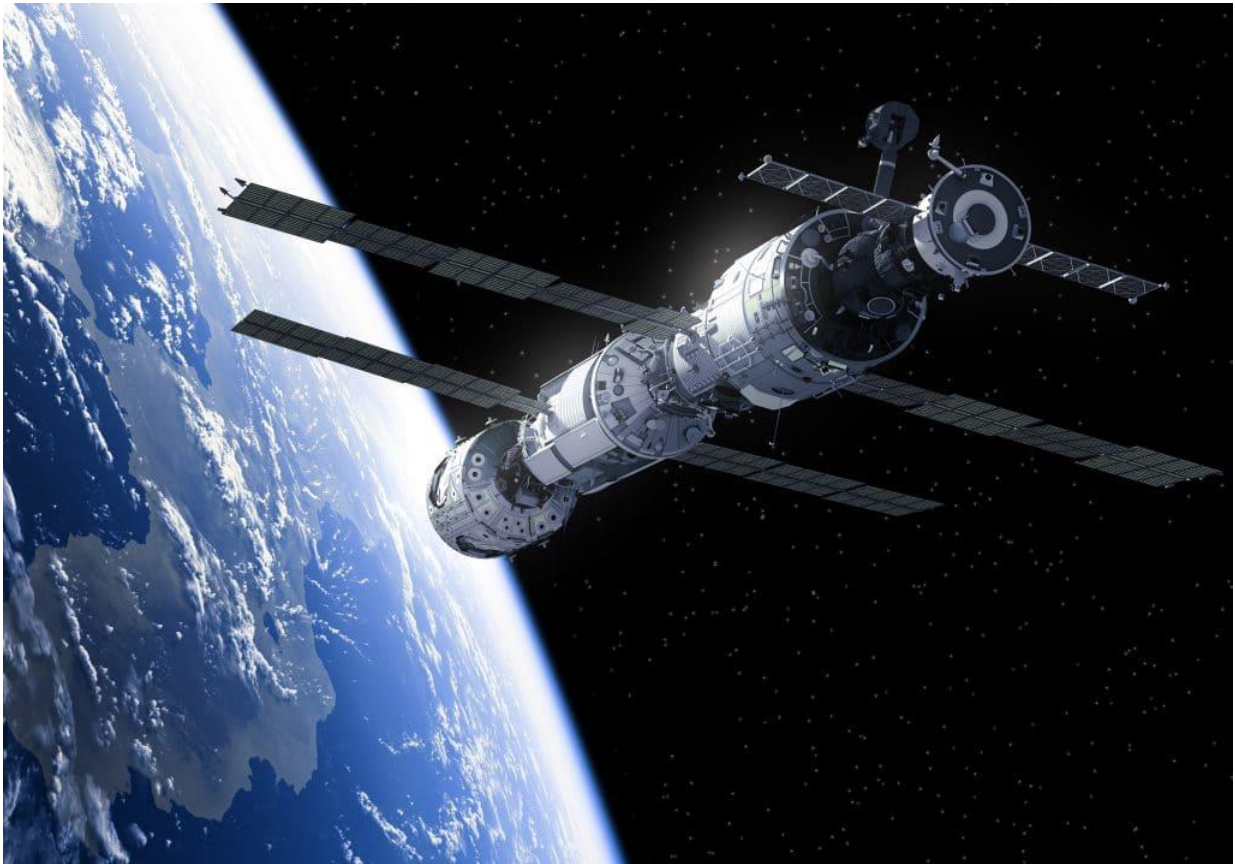


Policy Proposals for Cosmic Quarry Mining Operations



Chloe Seu
Joy Zhang
Lachlan Liu
Maria Jang
Min-Suh Park

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1. Executive Summary

This report assesses the key risks faced by Cosmic Mining Corporation across three solar systems: Helionis Cluster, Bayesia System and Oryn Delta, and provides recommendations of Galaxy General Insurance Company's product portfolios across four hazard areas: Equipment Failure, Cargo Loss, Workers' Compensation, and Business Interruption. Based on the risk analysis and scenario testing, the team recommends that Galaxy offer tailored insurance products for all four hazard areas in its portfolio, as each hazard represents a significant and distinct risk exposure. Business Interruption and Cargo Loss were identified as the most financially significant risks due to their potential to generate large losses and operational shutdowns affecting Cosmic Mining Corporation's inter-system transportation and supply chain reliance. Equipment failure represents a material risk given the high replacement costs of specialised machinery and the potential for production delays while Workers' Compensation presents a stable but necessary line of coverage due to ongoing workforce exposure.

Each recommended product includes flexible design features such that coverage limits, premiums and risk loadings can be tailored to reflect the different environmental conditions, operational scale and risk profiles present in each solar system. These include risk-index based premium adjustments, flexible coverage limits depending on asset values and operational scale and optional add-ons such as catastrophic event cover and supply chain disruption cover. This allows Galaxy to price risk appropriately while ensuring Cosmic Quarry Mining Corporation is adequately protected across all locations.

The recommended products are also designed to be adaptable and scalable, allowing pricing and coverage structures to adjust over time in response to future changes in technology, operational expansion, inflation and environmental conditions. These products can be integrated into Cosmic Quarry's enterprise risk management framework by transferring high severity risks, stabilising financial outcomes and supporting long-term operational planning across multiple solar systems.

2. Product Design

Premiums are paid to Galaxy General Insurance Company annually at the beginning of each year, starting from 2175. Claims are settled annually at the end of each year, with claims arising in 2175 paid in January 2176. The short-term policy has a duration of one year, while the long-term policy spans ten years, with final claims paid in January 2186.

2.1 Cargo Loss

The cargo loss product follows an indemnity-based benefit structure, where the insurer reimburses the value of cargo lost or damaged, up to the declared cargo value, ensuring claims are capped to prevent overcompensation. Cargo loss insurance covers goods transported via road, sea, air and rail with payouts triggered by loss, damage or delay from unforeseen events during transit. Key triggers include transport accidents such as collisions and overturning, particularly on high debris routes, as well as theft, hijacking and piracy for high-value cargo and high-risk routes (GSK Insurance, 2023). Environmental triggers include severe weather (e.g. floods, bushfires) and exposure to extreme conditions such as high solar radiation. Coverage also extends to accidental damage during handling, including loading and unloading of cargo. The policy excludes losses arising from improper packaging or storage, as these are preventable and are due to the failure to adhere to safe practices. This also

excludes litigation expenses where the conveyance is unauthorised to drive or is driving under the influence (National Transport Insurance, 2023). A flexible risk factor is applied to each solar system during pricing, allowing the product to adapt as risk levels change over time. Bayesia System has the lowest risk due to reliable communication and well-mapped asteroid routes. This is followed by Helionis Cluster, with moderate risk from asteroid clusters and navigation challenges. Oryn Delta has the highest risk due to weak signals and low visibility which directly threaten communication and safe transportation. The product uses pricing parameters to ensure scalability based on expected cargo volume by solar system, while maintaining adaptability to adjustments in navigation systems, safety features and environmental conditions through the risk factor.

2.2 Equipment Failure

To protect against the financial impact of mechanical and operational breakdowns across mining equipment, the product is designed as an indemnity-based cover. Given the modelling of equipment-related claims showing high frequency with moderate severity, the benefit structure covers repair and replacement costs subject to policy limits and deductibles to manage attritional losses. A tiered coverage structure is incorporated to offer varying levels of protection, ranging from basic repair only cover to comprehensive replacement and downtime cover. Coverage is triggered by unforeseen and sudden equipment failure during operation with triggers linked to key risk drivers used in modelling. Coverage is subject to compliance with prescribed operational standards differentiated by solar systems such as radiation hardening certified equipment in the Bayesia system and reporting gravitational shear zones in Oryn Delta. Deductibles are structured to reduce the frequency of minor claims on Helionis Cluster where debris-related damage is much more frequent. This structure allows the policy to be tailored to operational differences across solar systems and varying risk profiles.

Exclusions apply to losses arising from inadequate or delayed maintenance, operational misuse or operating outside of implied thresholds, or gradual wear and tear, ensuring consistency with underwriting assumptions and mitigating moral hazard. Severe stellar flare events or large-scale gravitational disturbances are also subject to exclusions or coverage limits. To reflect the differing operational environments across the Helionis Cluster, Bayesia system and Oryn Delta, product parameters on pricing are derived from these system-specific risk profiles whilst also incorporating exposure-based pricing linked to the quarry's equipment volume and productivity growth to account for the company's plans for enlarged scalability. Adaptability through flexible index factors can be adjusted in response to evolving technological advancements, maintenance standards and environmental conditions.

2.3 Worker Compensation

The workers compensation product provides comprehensive protection against accidents, injuries, and illnesses which arise out of and in the course of employment, consistent with established workers compensation systems (Knoblauch, 2023). With the support of automated technology like Interplanetary Claims Grid and Q-RISK Engine for efficient claims management and accurate risk assessment, the benefit structure is based on the extent of incapacity and recovery requirements, ensuring flexibility across a wide range of injury outcomes. This includes full coverage of medical, hospital and rehabilitation expenses, weekly wage replacement proportional to lost earning capacity during periods of incapacity, and additional compensation for permanent impairment or fatal outcomes

(SIRA, 2025). Coverage is triggered when an injury or illness arises out of employment, including workplace accidents, hazardous environmental exposure, and operational incidents, reflecting the standard no-fault principle underpinning workers' compensation schemes (Knoblauch, 2023). To mitigate cases of moral hazard, exclusions apply to non-work-related injuries, intentional harm, fraudulent claims, and incidents resulting from failure to comply with mandated safety protocols and meet a certain level of safety training and protective gear quality. A system-specific loading factor is applied to reflect differing operational risks across solar systems, with Helionis receiving the lowest loading due to established and predictable conditions, Bayesia System a moderate loading due to environmental stresses such as radiation and temperature extremes, and Oryn Delta the highest loading due to poor infrastructure and heightened uncertainty, allowing for recalibration with evolving hazard conditions, data documentation, and infrastructure. Claim losses are simulated based on the insurer's current company size, allowing the model to be recalibrated as the company grows, ensuring pricing remains accurate and proportionate as workforce scale and operational footprint increase over time.

2.4 Business Interruption

For the business interruption hazard area, a loss of gross profit benefit structure is being adopted, payments will be made throughout an indemnity period, closely reflecting the actual financial impact of disruptions. Given the extreme volatility of mining yields in the interstellar setting, this benefit structure was chosen as it provides flexibility and reduces the risk of material under/over compensation compared to fixed sum insured benefit structures. Coverage is triggered by defined interruption events that halt or materially reduce operations. Triggers can be categorised as internal hazards such as mechanical failure of critical extraction equipment, transport vessel breakdowns and external hazards such as meteorite impacts or solar radiation storms. Catastrophic incidents such as explosions stemming from equipment failure are also covered in this hazard area. However, the policy will incorporate exclusions to manage moral hazard and uninsurable risks. Similarly to the exclusions covered in equipment failure, claims cannot be made if halt in production is due to gradual wear and tear, lack of maintenance or known design defects. Interruptions caused by regulatory breaches, warlike or politically motivated acts and insolvency of the insured are also critical exclusions in this hazard area. Furthermore, coverage excludes losses stemming from predictable environmental conditions inherent to specific mining zones where adequate mitigation measures were not implemented.

3. Pricing and Capital Modelling

3.1 Aggregate Loss

Aggregate losses were modelled based on the distribution of frequency and severity in claims across the insurer's portfolio.

Equipment failure

Claim frequency was estimated using a Poisson model with exposure adjustments to reflect the risk profile in the quarry's inventory. These base frequencies were further adjusted for prospective conditions using scaled indices, capturing differences in both operational intensity and exposure growth especially in the Bayesia system and Oryn Delta relative to Epsilon. The productivity and environmental risk indices were applied multiplicatively to the modelled frequencies rather than modifying underlying coefficients. This approach ensures the integrity of the underlying statistical

model whilst maintaining flexibility in adapting to evolving risk and exposure. Claim severity was modelled using a lognormal distribution to reflect the right-skewed nature of loss amounts. The final frequency and severity models were selected by comparing AIC and BIC values across candidate models.

Cargo loss

Claim frequency was modelled using a Negative Binomial distribution due to overdispersion. Baseline frequency per unit exposure was scaled by total cargo capacity in the quarry inventory and adjusted using an exposure and risk index. The exposure index reflects each solar system's share of total capacity, while the risk index captures qualitative environmental and operational risks. These indices are flexible and can be updated as conditions change (Refer to Appendix 7.1 for detailed calculations). Claim severity was modelled using a lognormal distribution to capture extreme loss events, with model selection based on AIC and BIC comparisons. The resulting aggregate loss distribution was highly right-skewed, with the mean exceeding the median. (refer to Appendix 7.5 for cargo loss distribution) This indicated a low-frequency, high-severity pattern driven by a small number of extreme losses.

Workers' Compensation

Claim frequency was modelled using a Poisson distribution with exposure adjustments to reflect the risk profile for Cosmic Quarry Mining Corporation's current personnel. Baseline frequency per unit exposure was scaled by total quarry personnel and adjusted accordingly for each occupation. Claim severity was modelled using a lognormal distribution to model the heavily right-skewed claim severity. Both models were determined based on the lowest RMSE and MAE metric through k-fold cross validation.

Business Interruption

The claim frequency was modelled using a zero-inflated negative binomial distribution, as similarly with the other hazard areas the claims count data exhibited overdispersion. However, a zero-inflated model was necessary as the business interruption hazard area had lower claim frequency, resulting in an abundance of zero-claim observations. An offset term, defined as the logarithm of exposure, was incorporated into the model to account for differing levels of operational exposure across policies, ensuring that claim frequency is modelled on a per-unit exposure basis rather than raw counts. This allows for fair comparison between mining operations with varying durations and scales of activity. For claim severity, the log-normal distribution was selected due to its strong fit to positively skewed loss data and its ability to capture the heavy right tail associated with rare but high-impact losses. The gamma distribution was rejected due to its higher AIC value and lighter tails, while extreme value models like the generalised Pareto distribution were deemed unsuitable for the full dataset due to instability in parameter estimation and sensitivity to threshold selection.

Aggregate losses were simulated using Monte Carlo techniques with 100,000 iterations to generate full loss distributions for each hazard area. Claim severities were projected forward under a constant inflation rate, and the resulting annual losses were then discounted using appropriate short- and long-term interest rates. The following table displays expected losses with the 10th and 90th percentile representing the range. A perspective of the short and long term were provided, where short term is defined as the metrics observed in one year, whereas the long term is defined across the next decade.

Hazard	Term	Expected Loss	p10	p90	Variance	VaR99	TVaR99
Cargo Loss	Short	10,412.7	95.5	28,640.1	2.48e+14	75,022.6	96,115.7
	Long	101,089.3	41,011.7	278,045.6	2.34e+16	728,339.5	933,116.7
Equipment Failure	Short	84.8	81.0	88.6	8.84e+6	91.8	92.9
	Long	835.8	798.2	873.2	8.58e+8	905.1	915.8
Worker's compensation	Short	4.9	2.7	9.1	5.18e+7	37.8	55.1
	Long	47.4	26.3	88.5	4.88e+9	367.4	534.6
Business Interruption	Short	34,127.3	33,254.6	35,007.0	4.62e+11	35,712.8	35,951
	Long	361,832.2	352,579.6	371,158.6	5.19e+13	378,642.8	381,167.3

Figure 1: Expected Loss Distribution Summary per hazard. All values are in ₦ millions.

3.2 Returns and Net Revenue

Premiums were determined using a cost-plus pricing framework based on the simulated aggregate loss distributions. Using expected loss as the central estimate of claims cost, a risk margin was applied using hazard-specific multipliers on the hazard's standard deviation to reflect differences in volatility and tail risk across the portfolio. Higher margins were assigned to areas displaying greater severity and uncertainty where 2.0 was applied for cargo loss, 1.75 for business interruption, 1.5 for workers' compensation, and 1.25 for equipment failure. Further expense and profit loadings of 20% and 8% respectively were applied consistently on all product pricings (Reference Appendix 7.1 for detailed formulas). These assumptions are broadly aligned with industry practices, where higher-risk portfolios require greater capital buffers and risk compensation, while expense and profit loadings reflect typical operational cost structures and target returns for general insurers covering the intensive industrial industry.

Hazard	Term	Returns	Net Revenue
Cargo Loss	Short	58,217.8	36,161.5
	Long	565,193.6	351,065.6
Equipment Failure	Short	122.9	13.6
	Long	1,211.7	133.6
Worker's compensation	Short	21.8	12.6
	Long	211.4	121.7
Business Interruption	Short	49,050.4	5113
	Long	520,053	54,210.4
Total	Short	107,412.9	41,300.7
	Long	521,476.1	405,531.3

Figure 2: Returns and Net Revenue per hazard. All values are in ₦ millions.

From the results above, it is clear to see that the hazard areas with a higher variance and hence higher risk tend to result in higher returns, with equipment failure and worker's compensation contributing less than 1% to the projected net revenue for both the short and long term. Despite this, most of the profits must be reinvested into the business to ensure enough capital is on hold to pay off future claims, this is further supported by the fact that the value at risk for those products have a significantly higher value at risk.

3.3 Stress Testing

The following stress scenario was designed to approximate 1-in-100-year events by aligning with the extreme tails of the loss distributions where a macroeconomic shock impacts loss severity through inflationary pressures and interest rate hikes. The macroeconomic condition was based on the average rates over 2171 to 2173, to reflect the historical period of highest inflation and interest rates. The percentage change in net revenue was significantly greater for the long term (8.21%) than the short term (0.38%), indicating higher sensitivity to macroeconomic conditions over longer durations. This is driven by inflation (5.28%) exceeding both interest rates (short = 4.56%, long = 4.07%), causing cashflows to grow and compound more over time. This is consistent with Figure 4 where the long-term scenario shows a larger increase in returns relative to costs, leading to higher net revenue, while short-term impacts remain minimal. (Refer to Appendix 7.2 for calculations for each hazard).

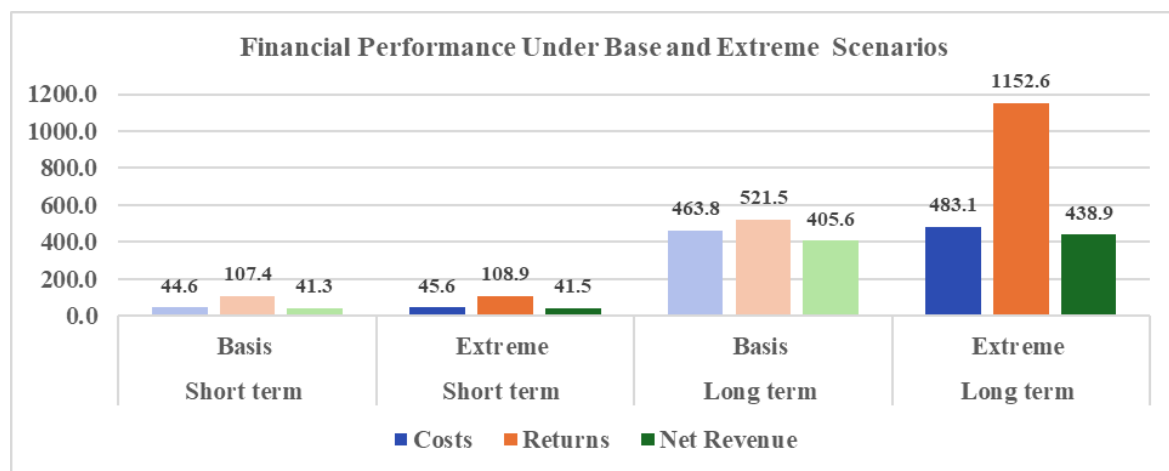


Figure 3: Financial Performance under Base and Extreme scenarios. All values are in D billions.

4. Assumptions

Assumption	Reasoning	Impact
Independent Claims Data	To keep the modelling and simulation process simple. More sophisticated methods to handle dependent data would only complicate the process without adding enough accuracy to our model's results.	Claim severity will be slightly mispriced as the model's underlying the business will be heavily bias towards the historical training data given.
Historical claim severity capped at the 99.5th percentile	Extreme values may be due to rare catastrophic events or data anomalies, which may distort the severity distribution and pricing if fully included. Allows the model to reflect typical large losses while limiting the influence of extreme outliers.	This may slightly underestimate the cost of extreme catastrophic losses, but results will be in more stable and realistic premium estimates for standard operations.
The relationship between risk factors and claim outcomes is assumed to be consistent across solar systems	Due to limited historical data for Bayesia System and Oryn Delta, data from similar systems (Zeta and Epsilon) was used as proxies and adjusted to reflect environmental risk differences when projecting claim losses.	If the true relationships differ significantly between solar systems, claim frequency and severity may be overestimated or underestimated, leading to potential mispricing for certain systems.

Expense, profit and risk margins are applied as a percentage loading to expected losses	Loadings are applied to ensure premiums cover operational costs, provide a return to stakeholders, and account for uncertainty in loss estimates. Industry benchmarks are used due to the lack of space mining insurance experience data.	Higher loadings increase premium levels but improve financial sustainability, while lower loadings increase the risk of under pricing and potential losses.
Risk index derived from quarry inventory data reflects underlying environmental risk in each solar system, independent of productivity risk.	Risk index as a proxy represents the environmental risks such as solar radiation asteroid activity and operating conditions in each solar system.	There may be errors in estimation of risk across solar systems as the index is a simplified representation of complex environmental and operational risks, potentially failing to capture all risk factors.

Other assumptions are listed in Appendix 7.3.

5. Risk Assessment

5.1 Key Risk Drivers

Helionis Cluster

Hazard	Risk	Definition	Mitigation
Cargo Loss	Cargo Volume Risk	Helionis has the highest concentration of cargo volume, increasing the likelihood of claims due to greater quantity and value of goods in transit	Adjust premiums to reflect the volume of cargo being transported, and set limits on the maximum insured capacity to control potential losses
Equipment Failure	Irregular gravitational resonances	Unstable gravitational interactions within a solar system caused by surrounding celestial bodies, accelerating equipment age and stressful usage intensity	Adjust premiums and underwriting controls based on proactive, rather than reactive, maintenance standards
Worker's compensation	Claims Risk	Increased risk of claims due to injuries caused by erratic drift patterns, micro-collisions, and shifting debris clouds.	Mandatory quantum-enhanced prediction models integrated into navigation systems to track shifting debris linked to underwriting eligibility
	Claims Delay & Escalation Risk	Delayed communication may lead to late reporting and escalation of injuries, increasing claim severity due to worsened medical conditions.	Expansion of the Interplanetary Claims Grid for real-time claims reporting via orbital AI adjudicators
Business Interruption	Rapid Spatial Cluttering	Communications system is prone to asteroid damage. Loss of communication may halt mining production increasing the change of business interruption.	Deploy multiple independent relay satellites in diverse orbits to prevent a single cluster fragmentation from cutting off all communication
Cargo Loss, Equipment Failure, Business Interruption	Asteroid Collisions	Asteroid collisions and debris clouds create navigation challenges, damage equipment quality, and wipe out multiple rigs or satellites.	Enforce protocols on low-risk route selection, verified equipment shielding systems and flexible modular operations with relocatable rigs

Bayesia System

Hazard	Risk	Definition	Mitigation
Equipment Failure, Cargo Loss, Business Interruption	Radiation damage	Binary star system leads to radiation spikes during orbital alignments, increasing claims on electronic damage and incidents during transportation. This may disable entire mining operations simultaneously leading to large losses in business interruption.	Sub-limits on radiation losses, with premium incentives for radiation-resistant equipment and low-exposure routing.
	Extreme temperatures	Extreme temperatures drive equipment failure, cargo degradation, and production downtime, increasing both claim frequency and severity.	Stricter durability and heat-resistant packaging requirements, with premiums adjusted for accelerated wear and temperature-related risks.
Worker's compensation	Catastrophic Claims Risk	Exposure to elevated ambient radiation and temperature extremes results in extreme-severity claims, including long-term disability and chronic illness benefits.	External risks are hard to predict so it is preferable to cede the catastrophic risk to a reinsurer. These claims can have a severe impact on profitability.
	Claims Accumulation Risk	Working extended periods in a high-gravity environment generates more frequent claims, leading to extended wage replacement lengths.	Underwriting control for a mandatory cap on hours worked per week to limit exposure accumulation

Oryn Delta

Hazard	Risk	Definition	Mitigation
Cargo Loss	Navigation instability	Weak signals disrupt communication with towers, reducing navigation reliability and increasing delays or cargo loss	Premium incentives for operators using high-reliability routes, with loadings applied to routes with historically weak signal coverage
	Low visibility conditions	Environmental factors such as fog and dust reduce visibility, impairing navigation and increasing accident risk	Strict underwriting controls requiring transportation vehicles to be equipped with advanced sensors such as radar
Equipment Failure, Business Interruption	Unpredictable stellar flares	Sudden and irregular bursts of stellar radiation can damage equipment	Real-time monitoring systems to detect flare events and adjust pricing models early Coverage limits for extreme flare events
	Gravitational shear zones	Regions where uneven gravitational forces lead to structural instability and increased failure risk	Regular reporting requirements on reinforced equipment for shear zones
Worker's compensation	Future Catastrophic Claims Risk	Increased risk of extreme-severity claims, due to future expansions into the asymmetric asteroid ring, where rapid orbital shear and fluctuating gravitational gradients.	External risks are hard to predict so it is preferable to cede the catastrophic risk to a reinsurer. These claims can have a severe impact on profitability.

	Rapid Infrastructure Development	Higher likelihood of claims due to rapid expansion of infrastructure indicating increases in short-term operational instability	Underwriting control for mandatory quality safety training and protective gear prior to worker deployment
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5.2 Ranking Top Threats

The risk matrix ranked the top threats:

1. Asteroid and Debris Collision
2. Radiation Exposure and Stellar Flares
3. Catastrophic Claims Risk
4. Gravitational Instability and Shear zone
5. Communication Instability

Figure 4 assigns rank by assigning a greater weighting to impact than likelihood.

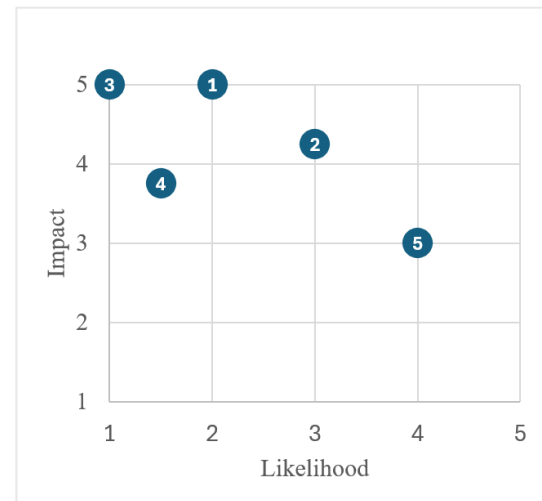


Figure 4: Risk Matrix for Top Ranking Threats

5.3 Correlated Risk Scenarios

In addition to individual hazard risks, correlated risk scenarios were considered where single events may result in losses across multiple hazard areas and/or solar systems simultaneously. A key correlated risk scenario is a solar storm which causes radiation and debris surges, navigation failures and communication disruptions, leading to equipment failure, cargo loss, workers' safety at risk and business interruption losses at the same time. The impact of this scenario is expected to vary across solar systems, with Bayesia likely to experience the greatest impact due to radiation spikes associated with its binary star system. On the other hand, Oryn Delta may experience significant communication disruptions due to its low visibility environment and reliance on relay systems which Helionis is expected to be affected the least due to its relatively stable solar environment, although debris-related disruptions may occur. Another scenario is major supply chain disruption arising, causing transport network failures, infrastructure breakdown or major operational incidents affecting interplanetary logistics. These may lead to cargo losses and business interruption while increasing equipment and worker-related risks due to delayed maintenance and operational stress. Given the reliance of mining operations on inter-related transport and communication networks across solar systems, such disruptions may lead to losses occurring across multiple areas simultaneously.

5.4 Scenario testing

To assess the sensitivity of profitability to key pricing assumptions, expense ratios were varied alongside the risk margin parameter k for each hazard area. The scenario testing showed a clear and consistent increase in net revenue as both the expense loading and risk factor k increased. Higher expense loadings directly raised premiums, leading to higher net revenue across all scenarios. Similarly, increasing k scaled the risk margin based on variability, resulting in higher premiums and therefore higher net revenue for all hazard areas. The percentage changes in net revenue (approx 98%) were broadly similar for both short-term and long-term policies, indicating a consistent sensitivity to these pricing assumptions across different time horizons. Individual calculations for each hazard can be found in Appendix 7.4.

Short Term	Expense Margin as % of Premiums				
Risk Factor	0.1	0.15	0.2	0.25	0.3
-0.5	775,990.3	780,726.9	786,121.4	792,321.0	799,520.6
-0.25	966,738.4	972,576.0	979,224.4	986,865.2	995,738.2
<i>k</i>	1,157,486.6	1,164,425.2	1,172,327.5	1,181,409.2	1,191,955.9
+0.25	1,348,234.7	1,356,274.3	1,365,430.6	1,375,953.4	1,388,173.5
+0.5	1,538,982.8	1,548,123.4	1,558,533.6	1,570,497.6	1,584,391.1

Long Term	Expense Margin as % of Premiums				
Risk Factor	0.1	0.15	0.2	0.25	0.3
-0.5	8,203,260.1	8,253,286.4	8,310,260.9	8,375,739.0	8,451,778.0
-0.25	10,221,783.7	10,283,461.2	10,353,704.8	10,434,432.3	10,528,181.3
<i>k</i>	12,240,308.7	12,313,637.0	12,397,149.6	12,493,126.6	12,604,583.6
+0.25	14,258,832.7	14,343,811.8	14,440,593.3	14,551,820.8	14,680,986.8
+0.5	16,277,356.7	16,373,986.6	16,484,037.1	16,610,514.0	16,757,390.1

Figure 5: Short- and Long-Term sensitivity of Total Net Revenue to risk factor (*k*) and expenses.

5.5 Risk differences amongst solar systems

The three solar systems present different risk profiles due to environmental and operational differences. Helionis Cluster has relatively stable conditions but higher frequency risks due to asteroid debris and micro-collisions, leading to more frequent equipment, cargo and worker-related claims. Bayesia System presents higher severity risks due to radiation exposure, extreme temperatures and high gravity conditions, potentially resulting in more severe equipment damage and worker-related injuries. Oryn Delta faces higher business interruption and operational risks due to poor visibility, communication failures and navigation instability, which may lead to operational downtime and transport disruptions. (Refer to Appendix 7.6 for ESG Considerations)

6. Data Limitations

6.1 Data Limitations and Impact on Results

Model outputs were validated against observed claim frequencies and severity distributions. However, limitations remain as the historical datasets were drawn from a different set of solar systems than those being modelled (Helionis Cluster, Bayesia System and Oryn Delta), leading to potential structural differences, particularly under extreme conditions. The lack of clear quantitative distinctions between solar systems in quarry personnel meant an exposure index could not be constructed, so it was proposed that an arbitrary loading factor be applied with Helionis Cluster acting as the benchmark. Similarly, the historical cargo loss data lacked clear distinctions between solar systems, limiting the model's ability to capture structural differences beyond qualitative adjustments. Data cleaning was not expected to materially bias the results.

6.2 Use of AI

AI tools including large language models, were used to assist with debugging R code, validating model outputs and checking the appropriateness of results.

7. Appendices

7.1 Formulas

Formula for Premium Calculations

$$\text{Premiums} = \text{NPV Expected Loss} + \text{Risk Margin} + \text{Expense Margin} + \text{Profit Margin}$$

Where :

$$\text{Risk Margin} = k * \text{Standard Deviation of Expected Loss}$$

$$\text{Expense Margin} = 0.2 * \text{Premium}$$

$$\text{Profit Margin} = 0.08 * \text{Premium}$$

$$\text{Net Revenue} = \text{Premiums} - \text{Expected Loss} - \text{Expense Margin}$$

Formula for NPV of Expected Loss

$$\text{NPV of Expected Loss over } n \text{ years} = \sum_{t=1}^n \frac{\text{Loss} * (1 + \pi)^t}{(1 + i)^t}$$

Where :

$$\text{Loss} = \text{Expected Frequency} * \text{Expected Severity}$$

$$\pi = \text{Inflation rate and } i = \text{Interest rate}$$

Formula for Cargo Loss frequency prediction

$$\text{Expected Frequency} = \text{Total Volume} * \text{Baseline Frequency} * \text{Exposure Index} * \text{Risk Index}$$

Where (Use Quarry Inventory):

$$\text{Total Volume} = \sum^{Vessel} \sum^{Solar \text{ system}} \text{Maximum Volume} * \text{Quantity}$$

$$\text{Exposure Index for solar system } i = \frac{\sum^{Vessel} \text{Maximum Volume} * \text{Quantity for } i}{\text{Total Volume}}$$

7.2 Stress Testing Tables for each Hazard

All values are in ₪ millions.

Cargo Loss	Monetary	Costs	Returns (Premiums)	Net Revenue
Short Term	Basis	10,412.7	58,217.8	36,161.5
	Extreme	10,415.4	58,232.7	36,170.8
Long Term	Basis	101,089.3	565,193.6	351,065.6
	Extreme	110,254.5	616,436.4	382,894.6

Equipment Failure	Monetary	Costs	Returns (Premiums)	Net Revenue
Short Term	Basis	84.8	124.0	14.4
	Extreme	82.6	120.7	14.0
Long Term	Basis	835.8	1,221.8	141.7
	Extreme	874.2	1,278.5	148.5

Workers Comp	Monetary	Costs	Returns (Premiums)	Net Revenue
Short Term	Basis	4.9	21.8	12.5
	Extreme	4.9	21.8	12.5
Long Term	Basis	47.4	211.4	121.7
	Extreme	49.1	269.1	166.1

Business Interruption	Monetary	Costs	Returns (Premiums)	Net Revenue
Short Term	Basis	34,127.3	49,050.4	5,113
	Extreme	35,118.8	50,475.4	5,261.5
Long Term	Basis	361,832.2	520,053	54,210.4
	Extreme	371,957.5	534,605.8	55,727.1

Sum	Stress cases	Costs	Returns	Net Revenue
Short	Basis	44,629.7	107,412.9	41,300.7
	Extreme	45,621.6	108,850.6	41,458.8
Long	Basis	463,804.7	521,476.1	405,631.3
	Extreme	483,135.3	1,152,589.7	438,936.4

7.3 Other Assumptions

Assumption	Reasoning	Impact
Exposure unit measured per Earth year	Using a consistent exposure period allows claim frequency and severity to be modelled and compared across hazard areas and solar systems on a standardised basis.	If exposure periods differ, claim frequency estimates may be over or understated.
Constant inflation rate of 2.465%	A constant inflation assumption simplifies pricing model and allows future claim costs to be estimated without introducing additional economic volatility	This may slightly underestimate the cost of extreme catastrophic losses, but results will be in more stable and realistic premium estimates for standard operations.
Constant short-term and long-term interest rates of 1.790% and 2.896% respectively	Due to limited historical data for Bayesia System and Oryn Delta, data from similar systems (Zeta and Epsilon) was used as proxies and adjusted to reflect environmental risk differences when projecting claim losses.	If the true relationships differ significantly between solar systems, claim frequency and severity may be overestimated or underestimated, leading to potential mispricing for certain systems.
Constant currency exchange rate where £1.00 = USD1.00	Loadings are applied to ensure premiums cover operational costs, provide a return to stakeholders, and account for uncertainty in loss estimates. Industry benchmarks are used due to the lack of space mining insurance experience data.	Higher loadings increase premium levels but improve financial sustainability, while lower loadings increase the risk of under pricing and potential losses.
All workers and machinery at Cosmic Quarry Mining Corporation will be covered with policies	Ensures that the exposure base used in modelling reflects the total operational risk faced by Cosmic Quarry Mining Corporation across all solar systems.	If not all workers or machinery are insured, the exposure may be overstated, leading to higher projected premiums and expected losses than in practice.
Number of workers and machinery insured remains constant over projection period	Simplifies projections of claim frequency and expected losses as exposure growth or decline, which would overcomplicate the model, is not modelled.	If the company expands operations or reduces its workforce and machinery over time, actual claim frequency and total losses may differ from projections which may lead to potential under pricing or overpricing.
Productivity index affects claim frequency but not claim severity	Higher productivity indicates increased operational activity and exposure which will subsequently increase the likelihood of accidents or equipment failure. Yet it may not necessarily increase the cost per individual claim.	Total losses could be underestimated given that higher productivity can lead to more severe incidents due to increased operational pressure and large-scale operations.
Claim frequency and severity distributions remain stable over time	Historical claims data used to model frequency and severity distributions.	There could be a potential difference in pricing if there are significant changes in technology, safety measures, environmental conditions or operational processes as future claim patterns would then differ from historical data.
Catastrophic events not explicitly	Pricing is based primarily on large losses rather than rare catastrophic	If a major correlated catastrophic event occurs, actual losses may be

modelled in base pricing	events as there is limited historical data on such events.	significantly higher than expected, potentially impacting the insurer's profitability and capital requirements.
Variant risk margin calculated as a multiple of the standard deviation of aggregate losses	Allows the risk margin to reflect differences in volatility and exposure uncertainty across hazard areas.	While it is an appropriate risk margin, the selection of multiplier is based on judgement and industry benchmarks which may not reflect the true uncertainty of hazards.

7.4 Scenario Testing Tables for each Hazard

All values are in £ millions.

Net Revenue: Risk factor (vertical) vs Expense Loading (horizontal)

Cargo Loss

	Short Term	Expense Margin as % of Premiums				
		0.1	0.15	0.2	0.25	0.3
Risk Factor (k)	1.5	26,949.1	27,164.8	27,410.3	27,692.6	28,020.4
	1.75	31,271.3	31,511.9	31,785.9	32,100.9	32,466.6
	2.0	35,593.5	35,859.1	36,161.5	36,509.1	36,912.7
	2.25	39,915.7	40,206.2	40,537.1	40,917.3	41,358.8
	2.5	44,237.9	44,553.4	44,912.6	45,325.5	45,805.0

	Long Term	Expense Margin as % of Premiums				
		0.1	0.15	0.2	0.25	0.3
Risk Factor (k)	1.5	261,629.2	263,722.8	266,107.2	268,847.5	272,029.7
	1.75	303,590.4	305,926.2	308,586.4	311,643.7	315,194.1
	2.0	345,551.5	348,129.5	351,065.6	354,439.9	358,358.4
	2.25	387,512.6	390,332.8	393,544.8	397,236.1	401,522.7
	2.5	429,473.8	432,536.2	436,023.9	440,032.2	444,687.1

Equipment Failure

	Short Term	Expense Margin as % of Premiums				
		0.1	0.15	0.2	0.25	0.3
Risk Factor (k)	0.75	10.7	11.3	11.9	12.6	13.5
	1.0	11.5	12.1	12.7	13.5	14.3
	1.25	12.4	12.9	13.6	14.3	15.1
	1.5	13.2	13.7	14.4	15.1	16.0
	1.75	14.0	14.6	15.2	15.9	16.8

	Long Term	Expense Margin as % of Premiums				
		0.1	0.15	0.2	0.25	0.3
	0.75	105.7	111.1	117.3	124.4	132.6
	1.0	113.7	119.2	125.4	132.6	140.9

Risk Factor (k)	1.25	121.7	127.2	133.6	140.8	149.2
	1.5	129.8	135.3	141.7	149.0	157.5
	1.75	137.8	143.4	149.8	157.2	165.7

Workers Compensation

	Short Term	Expense Margin as % of Premiums				
		0.1	0.15	0.2	0.25	0.3
Risk Factor (k)	1.0	8.4	8.5	8.5	8.6	8.8
	1.25	10.3	10.4	10.5	10.7	10.8
	1.5	12.3	12.4	12.5	12.7	12.8
	1.75	14.3	14.4	14.5	14.7	14.8
	2.0	16.3	16.4	16.5	16.7	16.9

	Long Term	Expense Margin as % of Premiums				
		0.1	0.15	0.2	0.25	0.3
Risk Factor (k)	1.0	81.3	82.0	82.9	83.9	85.0
	1.25	100.5	101.3	102.3	103.4	104.7
	1.5	119.6	120.6	121.7	123.0	124.4
	1.75	138.8	139.9	141.1	142.5	144.1
	2.0	158.0	159.1	160.5	162.0	163.9

Business Interruption

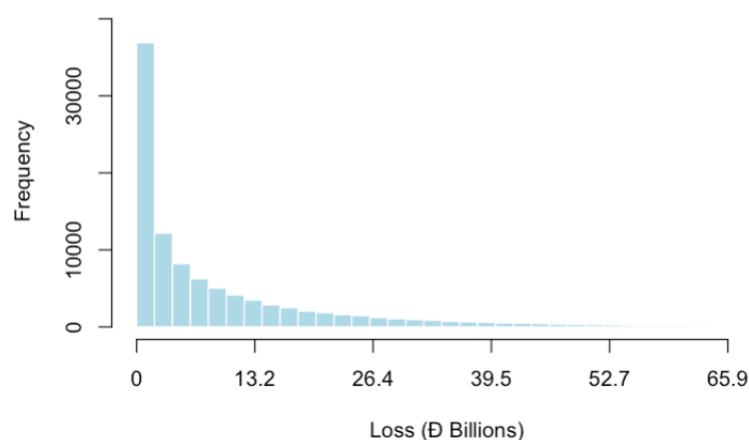
	Short Term	Expense Margin as % of Premiums				
		0.1	0.15	0.2	0.25	0.3
Risk Factor (k)	1.25	4,075.2	4,295.7	4,546.8	4,835.4	5,170.6
	1.5	4,261.6	4,483.2	4,735.5	5,025.6	5,362.4
	1.75	4,448.0	4,670.7	4,924.3	5,215.7	5,554.1
	2.0	4,634.5	4,858.2	5,113.0	5,405.8	5,745.9
	2.25	4,820.9	5,045.7	5,301.7	5,596.0	5,937.7

	Long Term	Expense Margin as % of Premiums				
		0.1	0.15	0.2	0.25	0.3
Risk Factor (k)	1.25	43,206.8	45,544.7	48,207.3	51,267.3	54,820.9
	1.5	45,183.4	47,532.7	50,208.3	53,283.2	56,854.1
	1.75	47,159.9	49,520.6	52,209.2	55,299.1	58,887.3
	2.0	49,136.5	51,508.6	54,210.1	57,314.9	60,920.5
	2.25	51,113.0	53,496.5	56,211.1	59,330.8	62,953.7

7.5 Predicted Loss Distribution

Cargo Loss

Aggregate Cargo Loss Distribution



7.6 Environmental, Social and Governance (ESG) Features and Considerations

In addition to the quantitative risk analysis, environmental, social and governance (ESG) factors were considered when assessing the overall risk exposure of Cosmic Quarry Mining Corporation's operations.

Under an environmental perspective, mining and transport activities can contribute to space debris, radiation exposure and environmental damage, which contributes to equipment failure and cargo loss risks over time. Implementing management and monitoring to reduce the likelihood of large losses and engaging in sustainable mining practices would help lower long term operational risks.

Social factors include workers' safety and wellbeing with a great emphasis placed given the hazardous and remote operating environments across the three solar systems. Fatigue, isolation and hazardous terrain can increase the likelihood and severity of workers' compensation claims. Compulsory safety protocols, protective equipment, training programs and access to medical support are essential to assisting with reducing risks.

Governance considerations involve strong risk management, safety compliance and incident reporting systems which are important for managing operational risks across multiple solar systems. Cosmic Quarry Mining Corporation has strong governance and safety frameworks, consequently expecting lower claim frequency and severity over time.

Overall, strong ESG practices support safer and more sustainable operations, which also reducing long term risk exposure.

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